

MAPLE CREEK, SK, Canada

WMO: 714530

Lat: 49.9025N

Lon: 109.4661W

Elev: 2515

StdP: 13.41

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -22.8	(c) -16.1	(d) -28.9	(e) 1.2	(f) -22.5	(g) -21.8	(h) 1.8	(i) -16.0	(j) 31.2	(k) 29.7	(l) 27.3	(m) 33.3	(n) 3.8	(o) 80	(p) 0.673

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.6	90.1	63.1	86.7	62.6	83.1	61.6	67.0	81.8	65.0	80.0	63.3	78.4	10.4	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.3	92.1	71.9	59.9	84.6	70.2	57.8	78.2	68.3	32.9	82.3	31.4	79.8	30.0	78.3	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a) 25.6	(b) 22.4	(c) 19.8	(e) -32.3	(f) 95.0	(g) 5.9	(h) 3.2	(i) -36.5	(j) 97.2	(k) -39.9	(l) 99.1	(m) -43.2	(n) 100.9	(o) -47.4	(p) 103.2
			(e) -32.5	(f) 70.3	(g) 5.8	(h) 2.9	(i) -36.6	(j) 72.4	(k) -40.1	(l) 74.1	(m) -43.3	(n) 75.8	(o) -47.5	(p) 77.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.7	17.0	19.4	30.2	41.9	51.6	59.5	67.0	65.2	55.7	42.7	29.5	19.1
	DBStd	21.04	17.00	15.35	14.42	9.32	8.30	6.39	6.20	6.85	8.78	9.75	13.48	15.77
	HDD50	4735	1022	858	616	270	80	6	0	1	39	265	619	959
	HDD65	8761	1487	1278	1078	693	423	188	49	83	300	693	1065	1424
	CDD50	1698	0	0	3	27	129	290	527	471	210	37	4	0
	CDD65	250	0	0	0	0	7	21	111	88	22	1	0	0
	CDH74	4022	0	0	0	10	138	376	1649	1418	415	16	0	0
	CDH80	1450	0	0	0	1	24	96	639	553	136	1	0	0
	WSAvg	8.6	8.3	8.1	9.5	9.9	9.5	8.4	8.1	7.8	8.0	8.9	8.8	8.4
	PrecAvg	17.2	0.9	0.7	1.0	1.4	1.9	3.5	2.0	1.8	1.5	1.1	0.8	0.8
(15) Precipitation	PrecMax	25.3	2.4	1.3	1.9	3.1	4.5	8.9	3.9	4.8	3.2	3.8	1.9	1.5
	PrecMin	8.6	0.2	0.1	0.1	0.5	0.1	1.3	0.3	0.0	0.0	0.1	0.2	0.0
	PrecStd	3.5	0.6	0.4	0.5	0.7	1.1	1.8	1.0	1.3	0.9	0.9	0.5	0.4
	PrecStd	3.5	0.6	0.4	0.5	0.7	1.1	1.8	1.0	1.3	0.9	0.9	0.5	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.3	54.6	65.0	74.6	83.4	88.3	93.9	94.1	89.5	76.5	64.7	54.2
		MCWB	43.3	42.4	48.0	53.4	59.2	63.2	64.7	63.5	60.8	56.0	48.3	41.9
	2%	DB	47.8	48.2	59.0	69.0	77.9	82.4	90.5	89.9	84.1	69.7	58.2	49.0
		MCWB	39.4	39.5	45.4	51.2	57.5	62.8	64.4	61.9	58.8	52.5	45.8	39.6
	5%	DB	44.0	43.7	54.0	64.1	73.5	78.1	87.1	86.5	78.7	64.1	53.3	44.6
		MCWB	37.7	36.8	42.5	48.2	55.2	60.7	64.2	61.6	57.5	49.8	42.9	37.2
	10%	DB	40.4	40.1	49.3	59.3	68.7	74.3	83.2	82.4	73.3	59.1	48.3	40.4
		MCWB	35.3	34.4	39.8	45.6	52.9	58.9	62.8	61.0	55.3	47.0	39.8	34.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.4	43.4	49.6	55.1	61.8	68.3	71.8	67.5	63.6	57.1	50.2	44.1
		MCDB	51.9	53.1	62.0	71.1	76.3	80.9	85.9	82.3	82.5	73.4	63.8	52.7
	2%	WB	40.4	39.9	46.1	52.1	58.6	64.7	68.1	65.4	60.8	53.5	46.4	40.3
		MCDB	46.7	47.2	57.6	67.0	73.9	76.9	83.5	81.0	79.2	67.3	57.2	47.5
	5%	WB	37.7	37.2	43.2	49.3	56.4	62.3	66.0	63.8	58.5	50.5	43.2	37.4
		MCDB	43.4	43.4	53.0	62.3	71.2	75.0	81.2	80.1	75.3	62.2	52.5	44.0
	10%	WB	35.4	34.6	40.4	46.6	54.0	60.0	64.2	62.2	56.3	47.9	40.1	34.8
		MCDB	40.6	39.7	48.6	57.6	66.4	71.9	79.2	78.5	71.3	57.8	47.5	40.4
(35) Mean Daily Temperature Range	5% DB	MDBR	21.8	21.4	21.1	23.5	24.0	22.7	26.6	26.9	25.4	23.0	21.1	20.8
		MCDBR	22.2	24.1	27.5	31.1	31.3	29.6	32.3	34.0	34.0	30.3	26.7	21.9
		MCWBR	15.9	16.7	16.1	16.8	15.1	14.1	13.0	13.2	14.8	16.3	16.2	14.9
	5% WB	MCDBR	22.1	23.9	26.4	29.5	27.8	26.8	28.4	28.2	30.8	28.2	25.8	21.8
		MCWBR	16.4	17.3	16.4	16.6	14.2	13.9	13.1	12.4	14.8	15.8	16.6	15.7
(40) Clear-Sky Solar Irradiance	taub		0.270	0.269	0.298	0.352	0.377	0.362	0.369	0.388	0.338	0.299	0.268	0.256
	taud		2.371	2.441	2.438	2.299	2.272	2.379	2.370	2.288	2.404	2.496	2.477	2.392
	Ebn at Noon		249	283	291	283	279	284	280	268	272	267	251	239
	Edh at Noon		20	24	29	37	40	36	36	38	30	23	18	17
(44) All-Sky Solar Radiation	RadAvg		406	727	1150	1550	1848	1987	2158	1756	1271	791	446	313
	RadStd		41	56	75	104	125	131	131	127	106	82	37	27

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page